Syllabus Edition
First teaching 2025
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Production Processes: Lean Production

Exam code: 0450, 0986 & 0264, 0774

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Lean production techniques

- **Lean production involves the minimisation of the resources used in production**
  - **Less time** is required as the production process is organised in the most efficient way
  - **Fewer materials** are used, as there is a focus on waste reduction
  - **Less labour** is used, as lean production is typically capital intensive
  - The **space required for production is reduced** as a result of just-in-time inventory management
  - A **small number of trusted suppliers** work closely with the business
- The use of lean production is likely to lead to a competitive advantage
  - Lower **unit costs** are achieved due to minimal wastage so **prices may be lower** than those offered by competitors
  - **Better quality** of output is likely as a result of supplier reliability and carefully managed production processes

1. Just-in-time inventory control

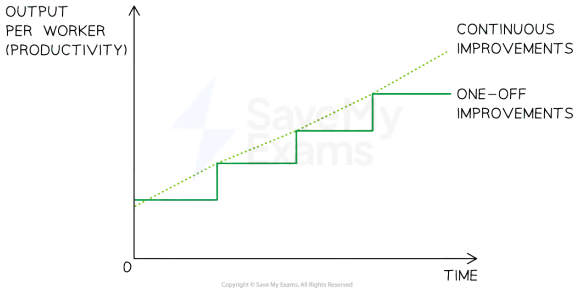
- **Just-in-time (JIT)** is a lean production technique where raw materials are **not stored on-site** but ordered as required and delivered by suppliers at the moment they are required for production
  - **Close relationships with suppliers** need to be developed to ensure reliable delivery and communication

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

2. Continuous improvement (Kaizen)

- Kaizen involves taking **continuous steps to improve productivity** through the **elimination of all types of waste** in the production process
  - **Changes are small and ongoing** rather than significant one-off's
  - They are **constantly reviewed** to ensure that they achieve the desired positive impact on productivity
  - Kaizen requires a **long-term management commitment** to change

Kaizen (continuous improvement)



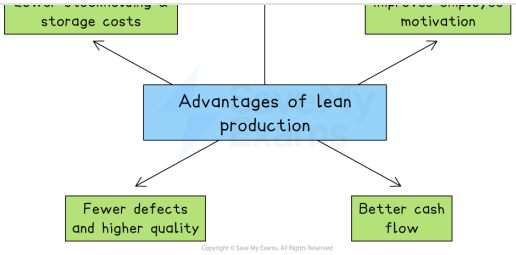
Kaizen Versus One-off Improvements

- Elements of Kaizen commonly include
  - **Zero defects** in manufacturing
  - High levels of **automation**
  - High levels of **cooperation** between workers and management
- Staff training and **computer inventory management systems** may also reduce wastage, as fewer errors are likely to be made

Help me understand

Evaluating the use of lean production

- Lean production has several advantages for businesses aiming to reduce waste and improve efficiency
- However, the initial implementation and ongoing monitoring of lean production has some significant drawbacks

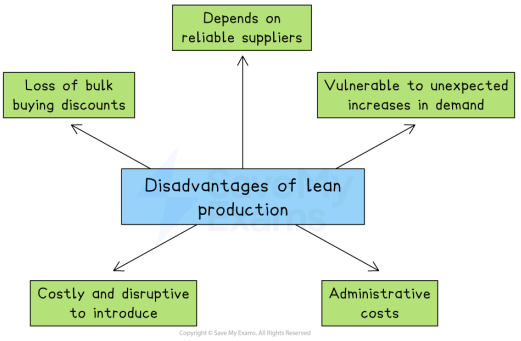


Lean production incurs low costs, improves employee motivation and should lead to higher quality

Advantages of lean production

- **Lean production helps to lower stockholding and storage costs**
  - Just-in-Time (JIT) systems reduce the need to keep large amounts of inventory
  - Storage space can be used for more productive activities, such as extra work areas or machinery
- **It encourages continuous improvement**
  - Employees regularly suggest small changes that help reduce waste and improve quality
  - This process allows problems to be identified and solved early
- **Employee motivation is likely to improve under lean production**
  - Workers are encouraged to work in teams and take responsibility for improvements
  - Being involved in decision-making can make staff feel valued and more committed
- **Lean production improves cash flow**
  - Because less money is tied up in stock, more cash is available for other business needs
  - This can help the business invest in areas like marketing, training, or technology
- **It can lead to fewer defects and higher product quality**
  - Lean methods focus on doing things correctly the first time to avoid waste
  - This results in more efficient production and greater customer satisfaction

Disadvantages of lean production



Lean production depends on reliable suppliers and makes the business vulnerable to sudden increases in demand

they may pay more for components and raw materials, increasing unit costs

- **It reduces the ability to respond quickly to unexpected increases in demand**
  - With little or no stock kept in reserve, businesses may miss sales opportunities
  - This can lead to dissatisfied customers or a damaged reputation
- **The system depends heavily on reliable suppliers**
  - If a delivery is late or poor in quality, production may come to a complete stop
  - Businesses must develop strong and dependable supplier relationships
- **Introducing lean production can be costly and disruptive**
  - It often requires big changes to the organisational structure and workflow
  - Employees may need extra training, which takes time and money
- **There are increased administrative costs due to frequent ordering**
  - Managing more orders creates extra paperwork and workload for staff
  - This can raise overall office and back-office expenses

Examiner Tips and Tricks

A common misconception is that lean production is always positive – strong answers recognise possible **drawbacks**, such as supply delays with Just-in-Time or employee resistance to continuous improvement

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